where $P_A(a)$ and $P_B(b)$ are the individual probabilities for a and b. (I added subscripts to the function symbols as a reminder that they could be different functions of their arguments.) It is easy to see that if the probability factorizes in this fashion, then there is no correlation; in other words, the average of the product is the product of the averages.

> **Exercise 6.1:** Prove that if $P(a, b)$ factorizes, then the correlation between a and b is zero.

Let me use an example to illustrate the kind of situation that leads to factorized probabilities. Suppose that instead of a single Charlie, there are two Charlies—Charlie-A and Charlie-B—who have never communicated. Charlie-B mixes up his two coins and gives one to Bob—the other one is discarded.

Charlie-A does exactly the same thing except that he gives a coin to Alice instead. This is the type of situation that leads to factorized product probabilities with no correlation.

In classical physics we use statistics and probability theory when we are ignorant about something that is, in principle, knowable. For example, after mixing up the coins in the first experiment, Charlie could have made a gentle observation (a quick peek) and then let Alice and Bob have their coins. This would have made no difference in the result. In classical mechanics, the probability distribution $P(a, b)$ represents an incomplete specification of the system state. There is more to know—more that could be known—about the system. In classical physics, the use of probability is always associated with an incompleteness of knowledge relative to all that could be known.

A related point is that complete knowledge of a system in classical physics implies complete knowledge of every part of the system. It would not make any sense to say that Charlie knew everything that could be known about the system of two coins but was missing information about the individual coins.